

SMART MULTI-CROP DISEASE DETECTION AND YIELD PREDICTION USING CNN AND LSTM MODELS



PRESENTED BY :

AADITYA KUMAR KARNA (PUR080BCT001)
ANKIT KUMAR YADAV (PUR080BCT016)
AVISHEK KUMAR GUPTA (PUR080BCT020)
HARIOM RAJ CHAUHAN (PUR080BCT033)

Introduction

- **Crop diseases and environmental changes** can reduce productivity and affect farmers' decisions.
- **Smart Multi-Crop Disease Detection and Yield Prediction** uses **deep learning** for disease detection and yield prediction.
- A **gatekeeper + crop-specific CNN** detects diseases, while an **LSTM model** predicts crop yield from agricultural and environmental data.
- The system supports **early disease detection and decision-making**.

Problem Statements

- **Crop diseases and unpredictable environmental conditions** can significantly reduce crop productivity. Traditional disease identification often depends on manual inspection, which can be time-consuming and may delay proper treatment.
- There is a need for an **intelligent multi-crop system** that can automatically identify the crop, detect its disease using **CNN-based models**, and predict crop yield using **LSTM and agricultural/environmental data** to support timely and informed farming decisions.

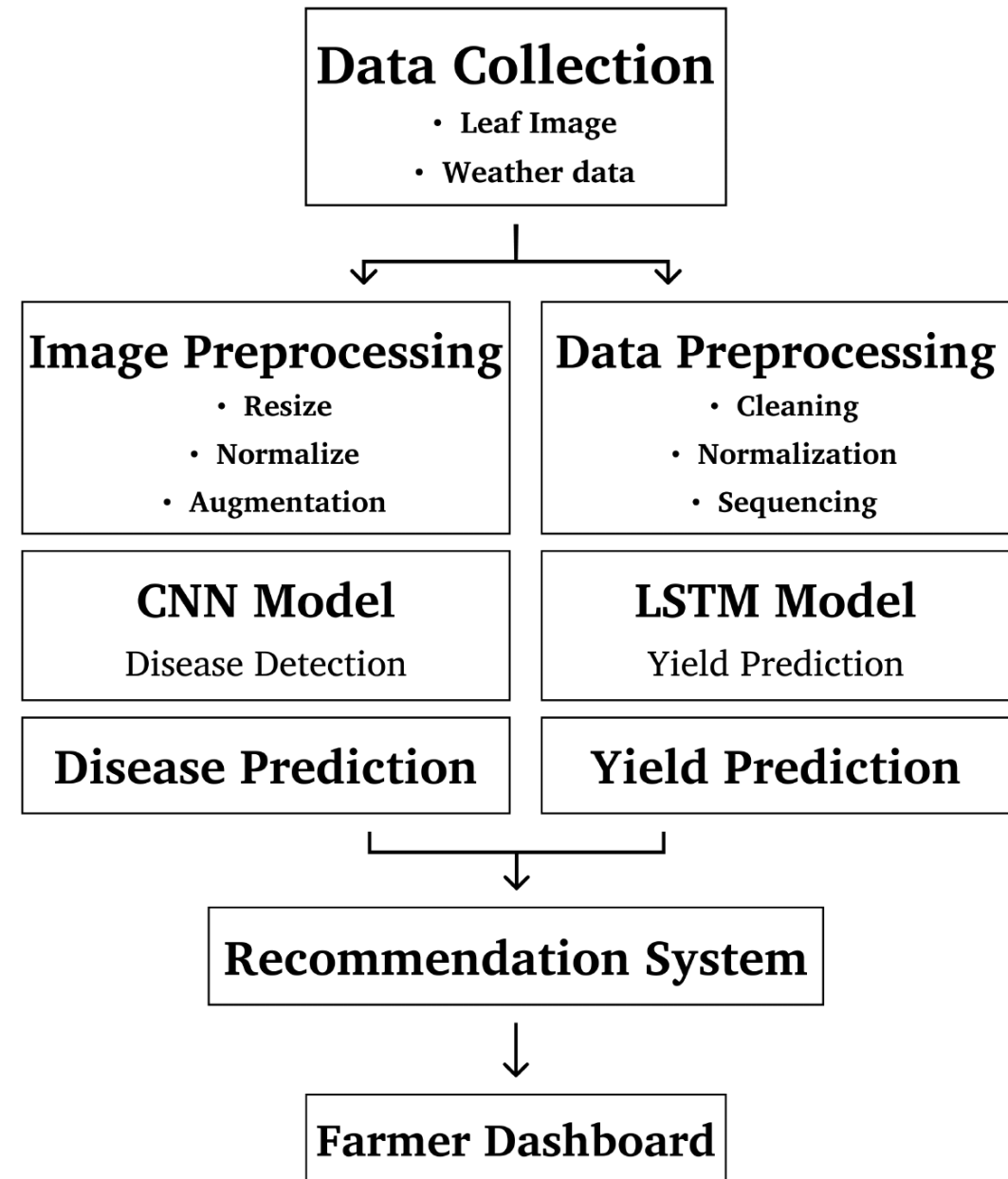
Objectives

- To design a **CNN and LSTM-based model** for detecting crop diseases and yield predictions from **leaf images** and environmental data.
- To integrate **disease detection** and **yield prediction** into a decision-support system for farmers.

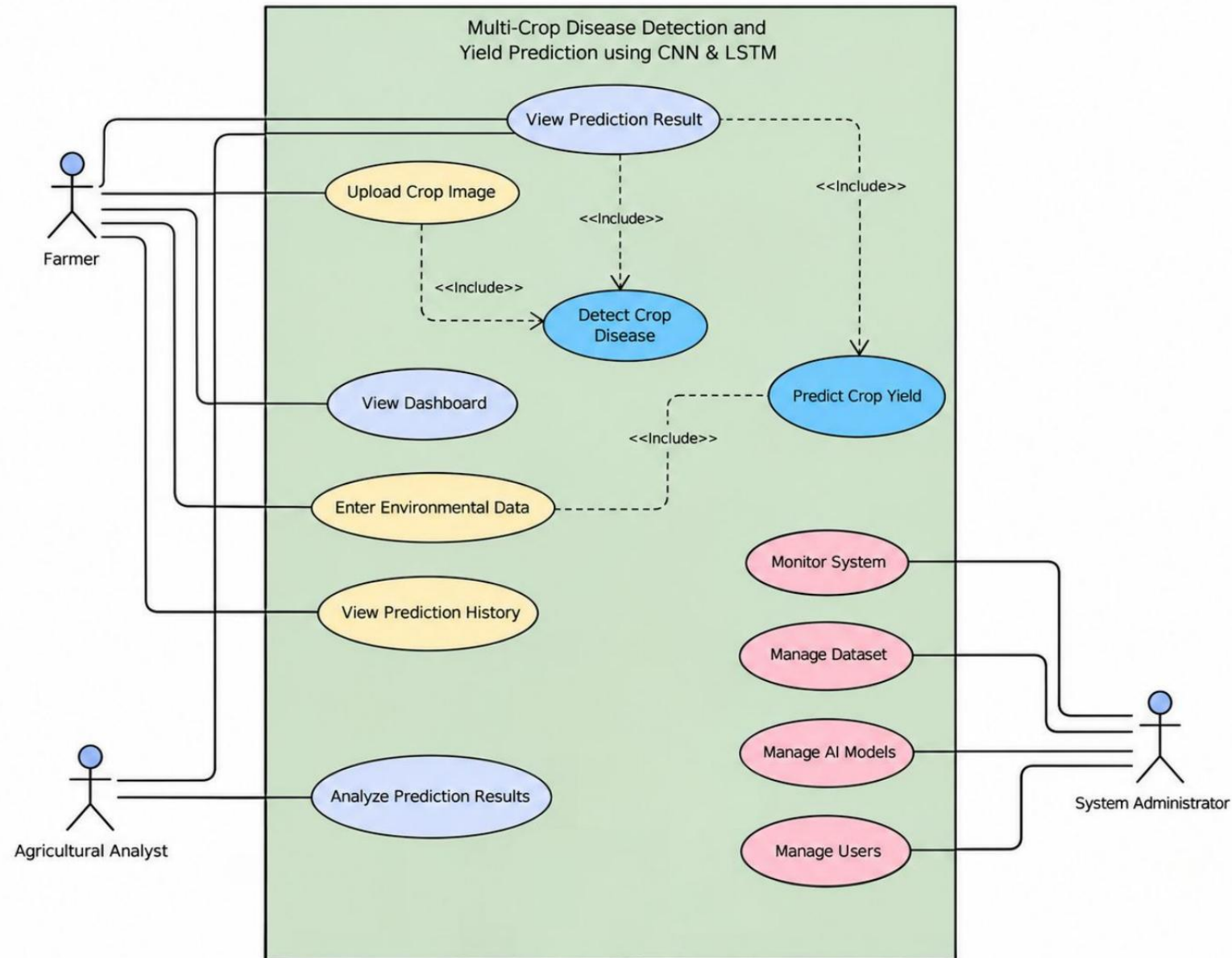
Literature Reveiw

- **(Akbar et al., 2022)** presented a CNN-based approach for crop disease detection, demonstrating the effectiveness of deep learning for identifying diseases from crop leaf images.
- **(Khaki et al., 2020)** proposed a CNN-RNN framework for crop yield prediction, showing that deep learning can effectively learn patterns from historical agricultural data.
- **(Liu et al., 2022)** explored hybrid CNN and LSTM models for smart agriculture, highlighting the benefit of combining disease detection with yield prediction.
- **(Too et al., 2022)** studied data augmentation techniques for plant disease classification and showed their usefulness in improving CNN model generalization.
- **(NARC, 2021) and Open-Meteo (2023)** provide agricultural/soil and weather data sources that can support crop analysis and yield prediction.

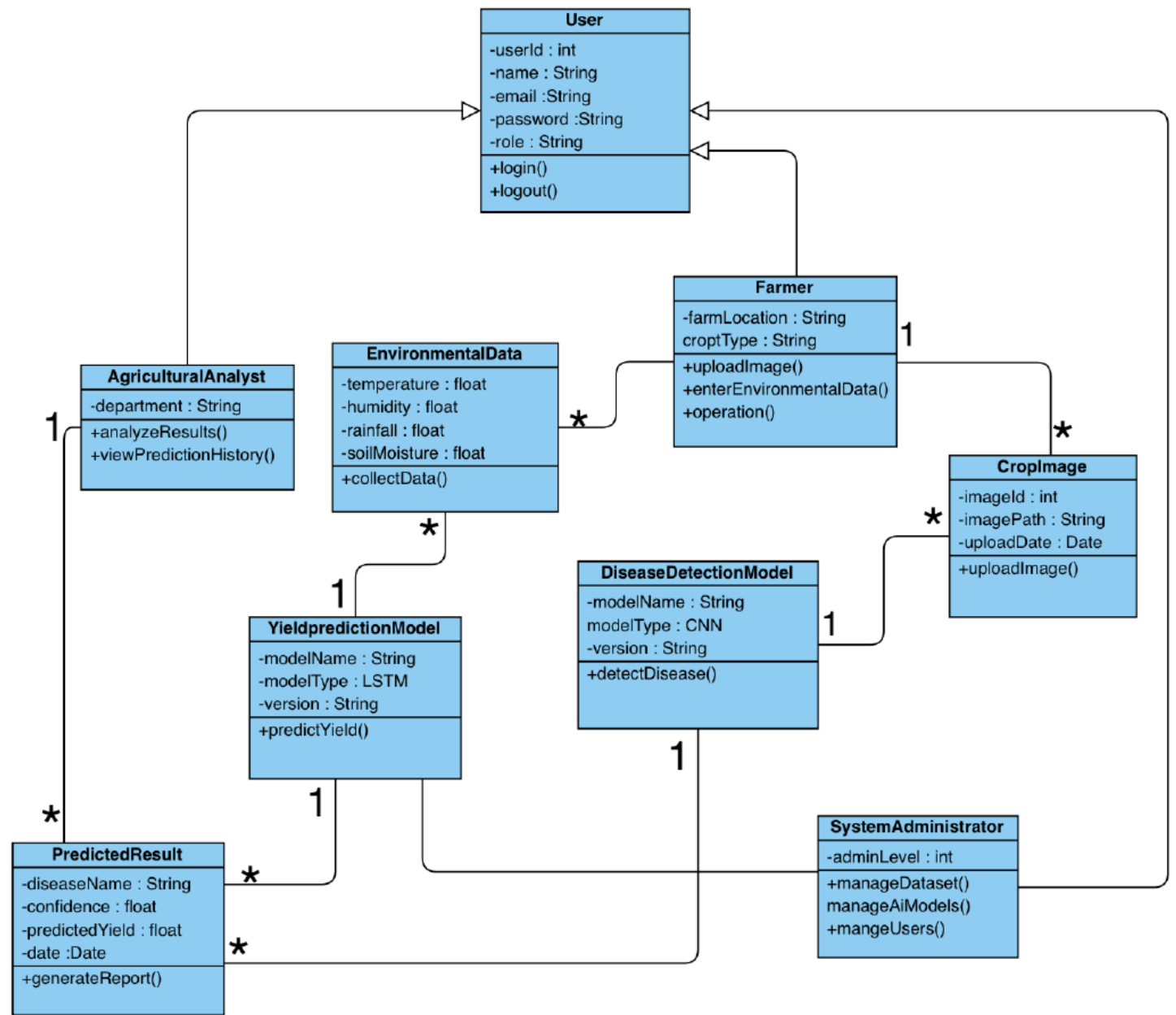
System Design



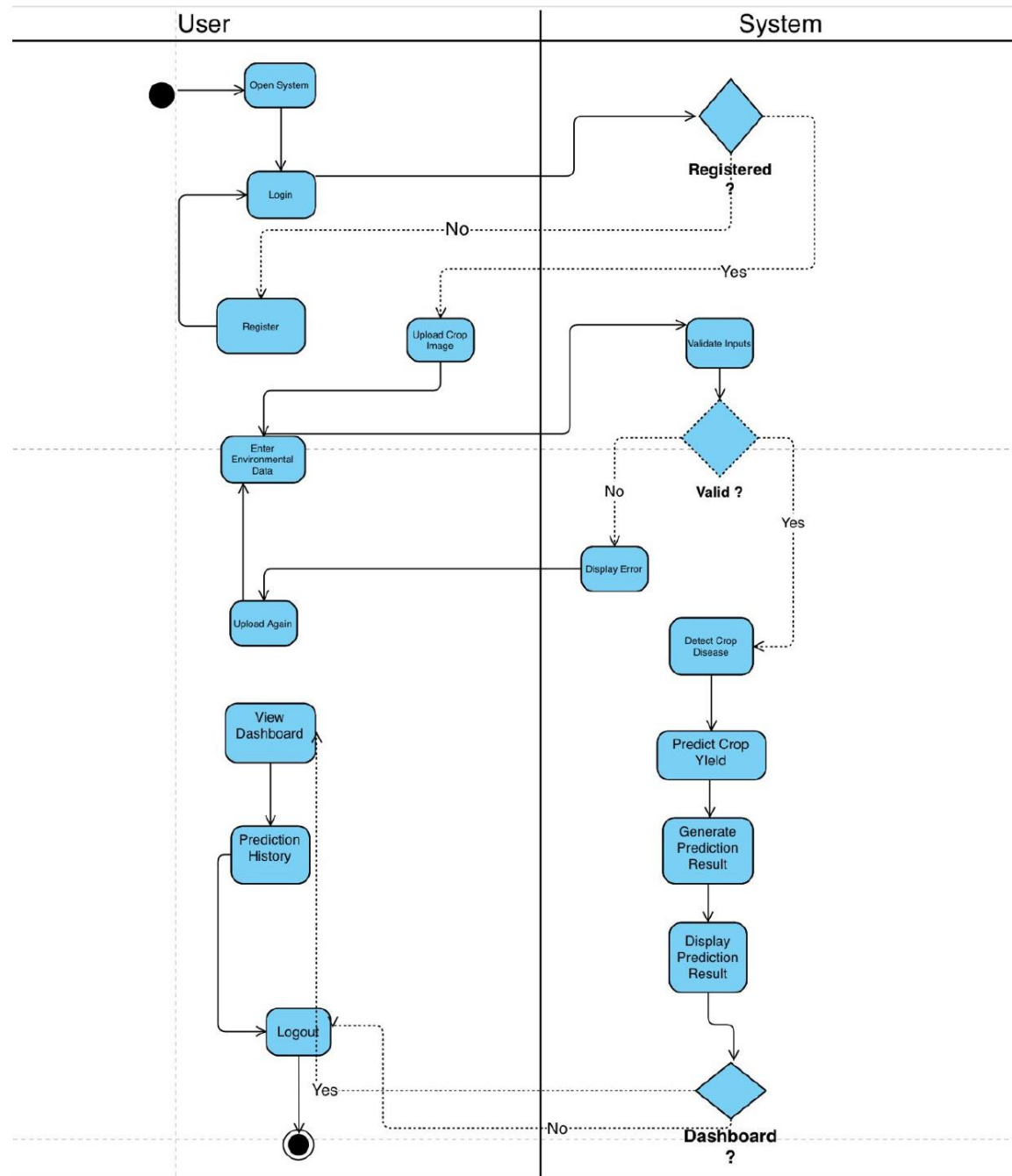
Use Case Diagram



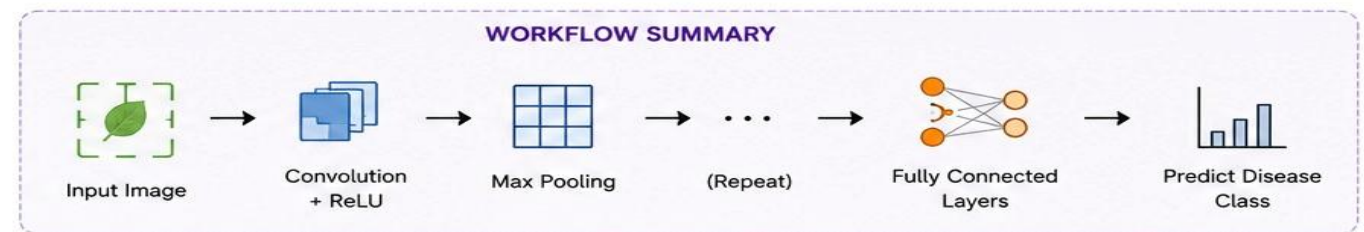
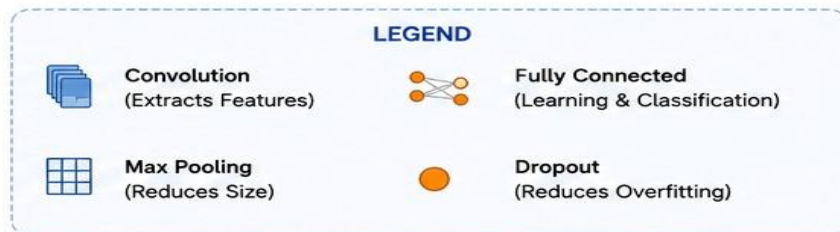
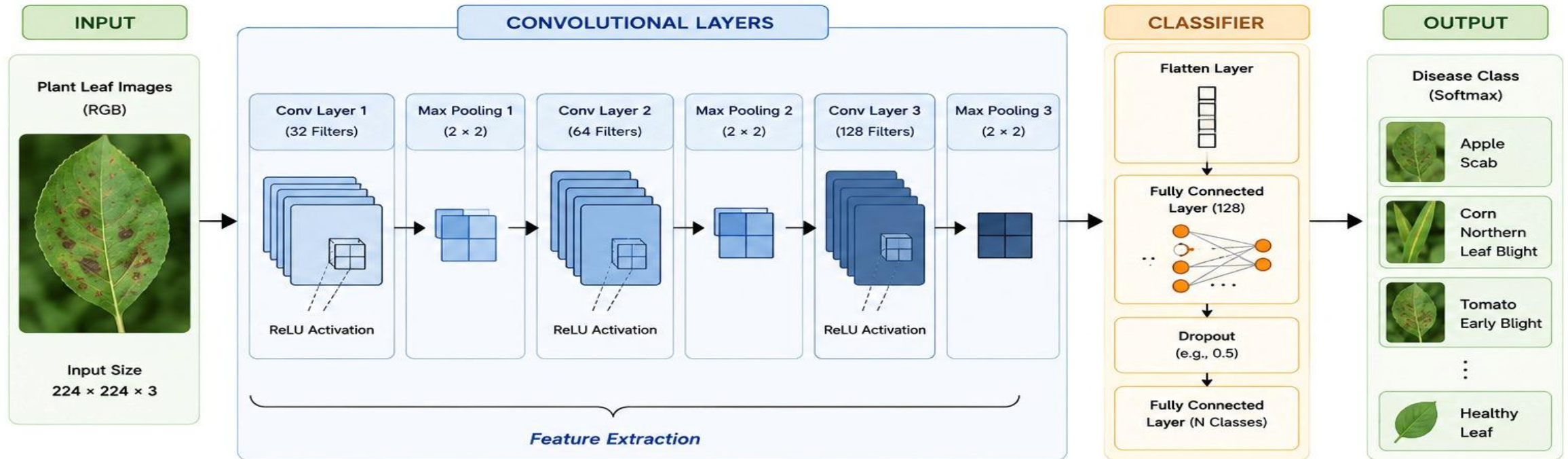
Class Diagram



Activity Diagram

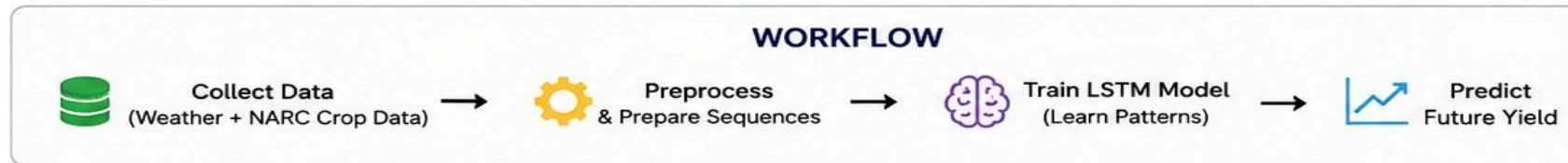
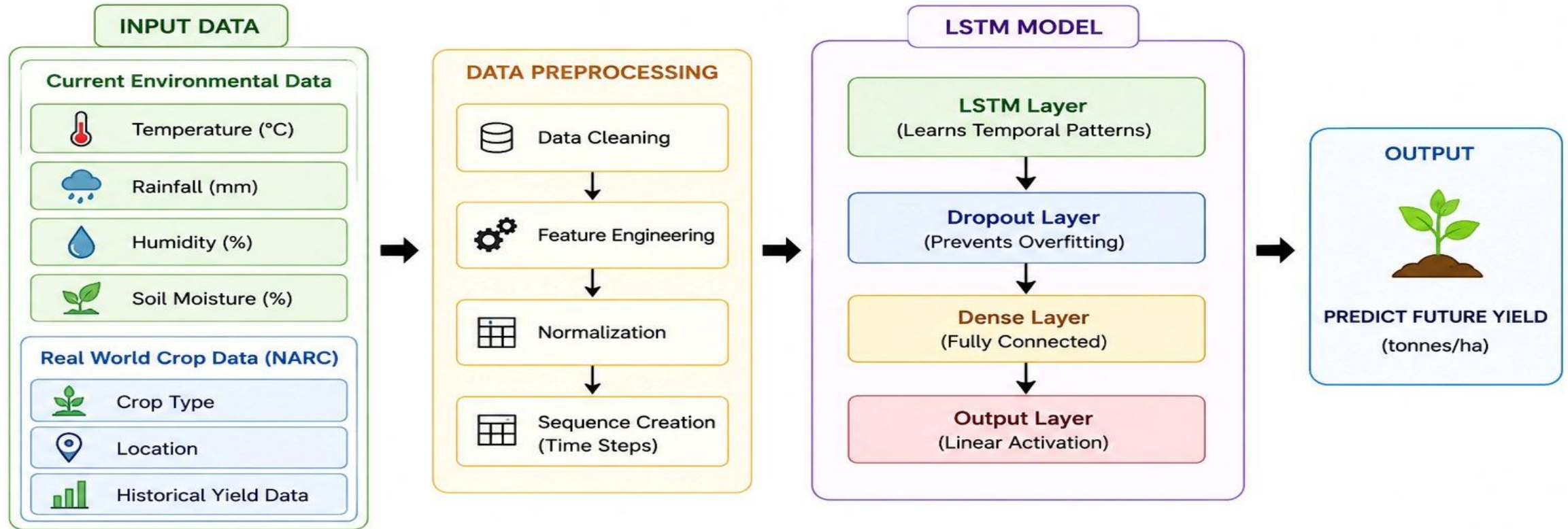


Simplified CNN Architecture for Plant Disease Detection



💡 This CNN automatically learns important features from leaf images and classifies the disease.

Simplified LSTM Architecture for Crop Yield Prediction



Dataset

Disease Detection Dataset

- Crop Type
- Disease Class / Image Label
- Leaf Images

Yield Prediction Dataset

- Rainfall, Temperature, Humidity
- pH value
- Historical Yield
- Soil & Environmental Conditions
- Weather Information
- Season / Timestamp
- Location

Implementation

- **Gatekeeper model** identifies the crop type from the uploaded leaf image.
- **Crop-specific CNN models** perform disease classification and provide confidence scores.
- **LSTM model** processes agricultural and environmental data to predict crop yield.
- **Web-based interface** connects the models and displays crop, disease, confidence, and yield results.
- **NARC data and Open-Meteo API** provide relevant soil, agricultural, and weather information for analysis.

Tools and Techniques

Tools

- Python
- Flask
- HTML/CSS/JavaScript
- VS Code
- Sqlite

Techniques

- Deep Learning
- CNN-Based Image Classification
- LSTM-Based Time-Series Prediction
- Data Preprocessing & Augmentation
- Real-Time Web-Based Prediction

Output

FC

Field Companion
Plant health + harvest forecast

DashboardSign in

1. Scan a leaf

2. Load field weather

3. Read the joined forecast

Leaf health

Upload a clear leaf photo. This CNN result is saved and later used by the yield buddy.

Crop

Orange

Leaf image

Choose JPEG, PNG, or WEBP



Scan leaf

Plant scan

Crop: orange

Finding: Orange__Haunglongbing(Citrus_greening)

Confidence: 100.00%

This scan found a problem. The yield buddy will lower the weather-only harvest number.

Orange__Haunglongbing(Citrus_greening)

100.00%

Future harvest

Use a Nepal agricultural region forecast or your own measurements. The spatial LSTM predicts weather yield; the buddy then adjusts it with plant health.

Crop for this forecast

Orange

Field region

Rupandehi

Load 12-step forecast

Loaded a 12-step future forecast for Rupandehi.

Temperature (°C)

29.5000

Rainfall (mm)

0.0000

Humidity (%)

85.0000

Soil moisture (%)

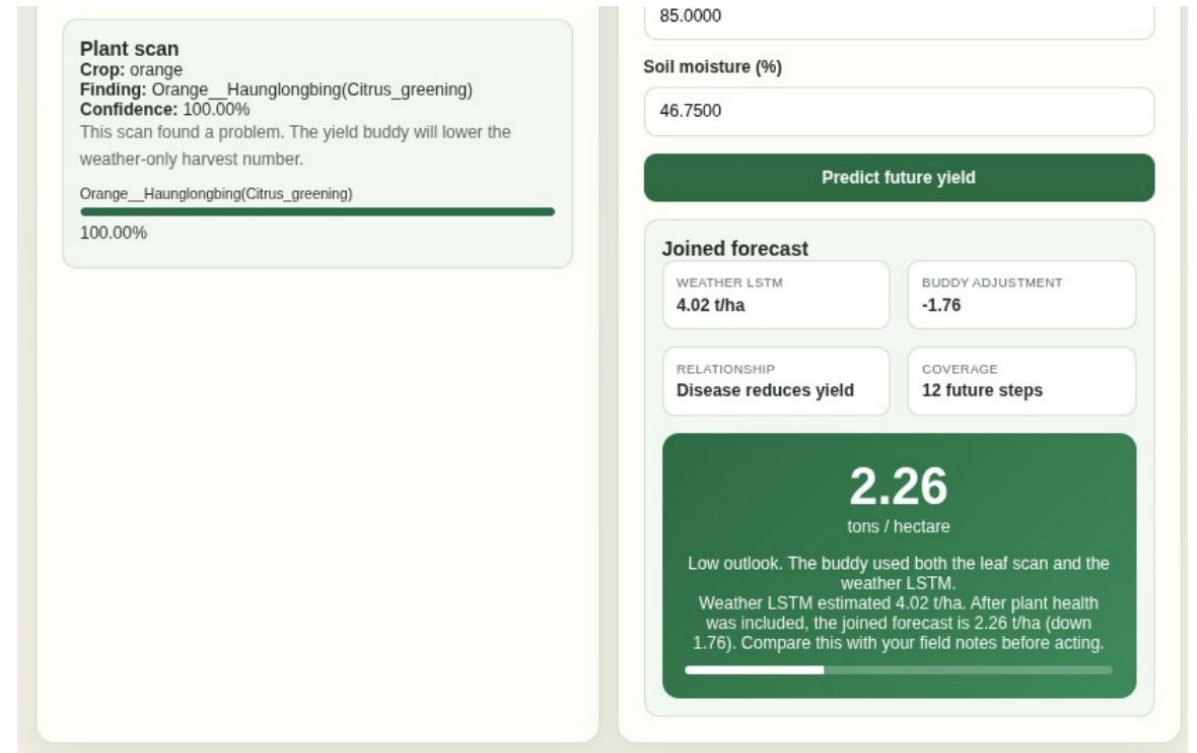
46.7500

Predict future yield

Joined forecast

15

Output



Application:

- **Helps farmers detect crop diseases early** using leaf image analysis and take timely action.
- **Predicts crop yield** using agricultural and environmental data to support better planning.
- Supports **crop monitoring, disease management, and informed decision-making** through a web-based intelligent system.

Limitation

- Depends on training data quality and diversity.
- Poor images or unseen diseases may reduce accuracy.
- Yield prediction requires accurate agricultural and environmental data.
- Supports only crops and diseases included in the trained models.

Future Enhancements

- Support more crops and disease classes.
- Integrate real-time weather and environmental data.
- Develop a mobile application for farmers.
- Provide AI-based disease treatment and crop recommendations.
- Improve model accuracy using larger and more diverse datasets.

References

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Any Queries?

Thank you!